

Chapter 3 Particles in the nucleus

Section 3.1 Atoms, isotopes and radioisotopes

Worked example: Try yourself 3.1.1

WORKING WITH ATOMIC NOTATION

How many protons, neutrons, nucleons and electrons are there in ${}_{92}^{252}\text{U}$?	
Thinking	Working
The lower number is the atomic number, Z . This is the number of protons.	atomic number, $Z = 92$ This nuclide has 92 protons.
The upper number is the mass number, A . This indicates the number of particles in the nucleus, i.e. the number of nucleons.	mass number, $A = 252$ This nuclide has 252 nucleons.
The number of neutrons, N , is the difference between the mass number, A (the number of nucleons), and the atomic number, Z (the number of protons).	$N = A - Z$ $= 252 - 92$ $= 160$ This nuclide has 160 neutrons.
In an electrically neutral atom the number of protons = the number of electrons.	The nuclide has 92 protons, so the atom will have 92 electrons in the electron cloud.

Worked example: Try yourself 3.1.2

WORKING WITH ISOTOPES

Consider the isotope of thorium, ${}_{90}^{230}\text{Th}$. Work out the number of protons, nucleons and neutrons in this isotope.	
Thinking	Working
The lower number is the atomic number, Z . This is the number of protons.	atomic number, $Z = 90$ This nuclide has 90 protons.
The upper number is the mass number, A . This is the number of particles in the nucleus, i.e. the number of nucleons.	mass number, $A = 230$ This nuclide has 230 nucleons.
Subtract the atomic number, Z , from the mass number, A , to find the number of neutrons, N .	$N = A - Z$ $= 230 - 90$ $= 140$ This isotope has 140 neutrons.

Section 3.1 Review

KEY QUESTIONS SOLUTIONS

- nucleons
- 79 protons and 118 neutrons ($N = A - Z = 197 - 79 = 118$)
- 235
- Chlorine-35 has 17 protons, 18 neutrons ($35 - 17$) and 35 nucleons.
 - Plutonium-239 has 94 protons, 145 neutrons ($239 - 94$) and 239 nucleons.

- 5 B and D. Carbon has 6 protons, so ^{13}C has 7 neutrons ($13 - 6$). Nitrogen has 7 protons, so ^{14}N has 7 neutrons ($14 - 7$).
- 6 The number of electrons in a neutral atom is the same as the number of protons, which is given by the atomic number.
- 7 Isotopes are atoms with the same number of protons but different numbers of neutrons.
- 8 **a** Their atomic numbers are the same as they are both krypton. Their mass numbers (84 and 89) are different as they are isotopes and have different numbers of neutrons.
b There would be no difference in their chemical interactions with other atoms.
- 9 A radioisotope is an unstable isotope. At some time, it will spontaneously eject radiation in the form of alpha particles, beta particles or gamma rays from the nucleus.
- 10 Yes, a natural isotope can be radioactive. For example, uranium is naturally occurring and every isotope of uranium is radioactive.
- 11 Since the nuclear strong force acts only over a short range, for larger nuclei more neutrons, compared to protons, are needed to balance the long-range electrostatic force.

Section 3.2 Radioactivity

Worked example: Try yourself 3.2.1

ALPHA DECAY

A radium-224 nucleus is known to decay to a new element through the emission of an alpha particle. Determine the new element, write its symbol and write the decay equation.	
Thinking	Working
From the periodic table, radium-224 has 88 protons. Therefore its atomic number, Z , is 88 and its mass number, A , is 224.	It can be written $^{224}_{88}\text{Ra}$.
The initial nucleus is $^{224}_{88}\text{Ra}$ and is written on the left-hand side of the equation. The unknown nucleus is a result of alpha decay (^4_2He) and is written on the right-hand side along with the alpha particle.	$^{224}_{88}\text{Ra} \rightarrow ^A_Z\text{X} + \alpha$ or $^{224}_{88}\text{Ra} \rightarrow ^A_Z\text{X} + ^4_2\text{He}$
Charge must be conserved, so the total number of protons, Z , must be the same.	$88 = Z + 2$ $Z = 86$
The number of protons and neutrons, A , must also be the same.	$224 = A + 4$ $A = 220$
For the new element, $Z = 86$. From the periodic table, this is radon.	$^{220}_{86}\text{Rn}$
The decay equation can now be written.	$^{224}_{88}\text{Ra} \rightarrow ^{220}_{86}\text{Rn} + ^4_2\text{He}$

Worked example: Try yourself 3.2.2
BETA-MINUS DECAY

An astatine-219 nucleus is known to decay to a new element through the emission of a beta-minus particle. Determine the new element, write its symbol and write the decay equation.	
Thinking	Working
From the periodic table, astatine-219 has 85 protons. Therefore its atomic number, Z , is 85 and its mass number, A , is 219.	It can be written ${}_{85}^{219}\text{At}$.
The initial nucleus is ${}_{85}^{219}\text{At}$ and is written on the left-hand side of the equation. The unknown nucleus is a result of beta-minus decay and is written on the right-hand side along with the beta-minus particle and an antineutrino.	${}_{85}^{219}\text{At} \rightarrow {}_Z^AX + {}_{-1}^0\beta + \bar{\nu}$.
Charge must be conserved, so the total number of protons, Z , must be the same.	$85 = Z - 1$ $Z = 86$.
The number of protons and neutrons, A , must also be the same.	$219 = A + 0$ $A = 219$
For the new element, $Z = 86$. From the periodic table, this is radon.	${}_{86}^{219}\text{Rn}$
The decay equation can now be written.	${}_{85}^{219}\text{At} \rightarrow {}_{86}^{219}\text{Rn} + {}_{-1}^0\beta + \bar{\nu}$

Worked example: Try yourself 3.2.3
RADIOACTIVE DECAY 1

After beta-minus decay from boron to carbon-12, the carbon-12 atom is in an excited state and decays further to a more stable form of carbon-12. The equation is ${}_{6}^{12}\text{C}^* \rightarrow {}_{6}^{12}\text{C} + X$. Determine the atomic and mass numbers for X and identify the type of radiation being emitted.	
Thinking	Working
Balance the mass numbers.	The mass numbers of 12 are already balanced, so the mass number of X is zero.
Balance the atomic numbers.	The atomic numbers of 6 are already balanced, so the atomic number of X is zero
X has an atomic number of zero and a mass number of zero.	X is a gamma ray, γ .

Worked example: Try yourself 3.2.4
RADIOACTIVE DECAY 2

Polonium-218 decays by emitting an alpha particle and a gamma ray. The nuclear equation is: ${}_{84}^{218}\text{Po} \rightarrow X + {}_2^4\text{He} + \gamma$. Determine the atomic and mass numbers for X , then use the periodic table to identify the element.	
Thinking	Working
Balance the mass numbers.	$218 = A + 4$ mass number = 214
Balance the atomic numbers.	$84 = Z + 2$ $Z = 84 - 2 = 82$
Use the periodic table to look up element 82.	Element 82 is lead.

Section 3.2 Review

KEY QUESTIONS SOLUTIONS

- 1 mass number of X is $218 - 214 = 4$
atomic number of X is $86 - 84 = 2$
 X is an alpha particle.
- 2 mass number of Y is $214 - 214 = 0$
atomic number of Y is $82 - 83 = -1$
 Y is a beta-minus particle.
- 3 beta-plus
- 4 A positron is a positively charged electron.
- 5 Alpha is a helium nucleus. Beta is a positively or negatively charged electron ejected from the nucleus. Gamma is electromagnetic radiation.
- 6 40, 42, 43, 44, 46, 48
- 7 Alpha, beta and gamma radiation all originate from the nucleus of an atom.
- 8 **a** X : atomic number = $92 - 2 = 90$, mass number = $235 - 4 = 231$, X is thorium
b Y : atomic number = $88 + 1 = 89$, mass number = $228 + 0 = 228$, Y is actinium
- 9 **a** Nitrogen-14 has 7 protons and 7 neutrons.
b A neutron has changed into a proton, an electron and an antineutrino.
- 10 **a** atomic number = $20 - 21 = -1$, mass number = $45 - 45 = 0$, beta-minus particle
b atomic number = $70 - 68 = 2$, mass number = $150 - 146 = 4$, alpha particle

Section 3.3 Properties of alpha, beta and gamma radiation

Section 3.3 Review

KEY QUESTIONS SOLUTIONS

- 1 **a** gamma
b beta minus
c alpha
d beta
e gamma
- 2 gamma
- 3 beta
- 4 gamma
- 5 beta and gamma
- 6 **a** nucleus
b nucleus
c nucleus
- 7 gamma, beta, alpha
- 8 Alpha particles travel through air at a relatively low speed and have a double positive charge, which means they readily ionise the air. Their charge, ionising ability and their relatively slow speed make them very easy to stop. This means that they have a very poor penetrating ability.
- 9 The wire should be a beta emitter, since the irradiation needs to be confined to a relatively small area. Alpha radiation does not have sufficient penetrating power, while gamma radiation would irradiate adjacent healthy cells.
- 10 Alpha particles will all be stopped by the metal sheet. Gamma rays will all penetrate the metal sheet. Differences in the thickness of the metal sheet will not affect the count rates of these two.
Some beta particles will pass through thin metal, so for a set metal thickness there is a set rate of beta particles that should make it to the other side.

Section 3.4 Half-life and decay series

Worked example: Try yourself 3.4.1

HALF-LIFE

A sample of the radioisotope sodium-24 contains 4.0×10^{10} nuclei. The half-life of sodium-24 is 15 hours. How many sodium-24 atoms will remain in the sample after 150 hours?	
Thinking	Working
Calculate how many half-lives 150 hours corresponds to.	$n = \frac{150}{15}$ $= 10 \text{ half-lives}$
Substitute $N_0 = 4.0 \times 10^{10}$ and $n = 10$ into the equation. Calculate the number of nuclei remaining.	$N = N_0 \left(\frac{1}{2}\right)^n$ $= 4.0 \times 10^{10} \times \left(\frac{1}{2}\right)^{10}$ $= 3.9 \times 10^7 \text{ nuclei}$

Worked example: Try yourself 3.4.2

HALF-LIFE AND ACTIVITY

A sample of strontium-90 has an initial activity of 4000 Bq. Calculate its activity after 6 months using Table 3.4.1.	
Thinking	Working
Determine the half-life of strontium-90.	$t_{1/2} = 28.8 \text{ years}$
Calculate how many half-lives 6 months corresponds to. Convert the half-life into months first.	$t_{1/2} = 28.8 \times 12 \text{ months}$ $= 345.6 \text{ months}$ $n = \frac{6}{345.6}$ $= 0.0174 \text{ half-lives}$
Substitute the initial activity, $A_0 = 4000$, and the number of half-lives, $n = 0.0174$, into the equation. Calculate the final activity.	$A = A_0 \left(\frac{1}{2}\right)^n$ $= 4000 \times \left(\frac{1}{2}\right)^{0.0174}$ $= 4000 \times 0.988$ $= 3950 \text{ Bq}$ Even after 6 months the activity has not significantly changed.

Section 3.4 Review

KEY QUESTIONS SOLUTIONS

- The activity is the count rate or the number of decays each second.
- $$N = 8.0 \times 10^{10} \times \left(\frac{1}{2}\right)^1$$

$$= 4.0 \times 10^{10}$$
- $$n = \frac{24}{8}$$

$$n = 3$$

$$N = 2.4 \times 10^{12} \times \left(\frac{1}{2}\right)^3$$

$$= 3.0 \times 10^{11}$$

- 4 a Halve successively from a starting number, e.g. 800, until 0.1% of 800 (0.8) is reached:
 $800 \rightarrow 400 \rightarrow 200 \rightarrow 100 \rightarrow 50 \rightarrow 25 \rightarrow 12.5 \rightarrow 6.25 \rightarrow 3.125 \rightarrow 1.56 \rightarrow 0.78$.
 This takes 10 halvings
 or
 $0.1\% = 0.001$
 $\left(\frac{1}{2}\right)^n = 0.001$
 take logs of both sides
 $n \log\left(\frac{1}{2}\right) = \log 0.001$
 $-0.3n = -3$
 $n = 10$
 It will take 10 half-lives to fall below 0.1%.
- b 10 half-lives must pass = $10 \times 24\,000 = 240\,000$ years
- 5 The percentage chance any atom has of decaying in a period of time equal to its half-life is always 50%.
- 6 number of half-lives = 4
 $12 = N \times \left(\frac{1}{2}\right)^4$
 $12 = N \times 0.0625$
 $N = 12 \div 0.0625 = 192$
 so 192 μg must be produced.
- 7 $6000 \rightarrow 3000 \rightarrow 1500 \rightarrow 750 \rightarrow 375$
 So 4 half-lives have passed:
 $60 \div 4 = 15$
 So the half-life of the radioisotope is 15 minutes.
- 8 a time to fall from $800 \rightarrow 400 = 10$ minutes or from $400 \rightarrow 200 = 10$ minutes
- b 40 minutes = 4 half-lives; $A = 800 \times \left(\frac{1}{2}\right)^4 = 50$ Bq
- 9 Lead-210 undergoes beta decay. Its half-life is 20 years.
- 10 Starting from uranium-234, seven alpha and four beta-minus decays have occurred.

Section 3.5 Radiation dose and its effect on humans

Worked example: Try yourself 3.5.1

ABSORBED DOSE

A cancer tumour is exposed to 0.50J of radiation energy. The absorbed dose is 3 Gy. Calculate the mass of the tumour. Assume that all of the radiation is absorbed by the tissue.	
Thinking	Working
Rearrange the equation $AD = \frac{E}{m}$.	$m = \frac{E}{AD}$
Substitute the values into the equation and solve.	$m = \frac{3}{0.5}$ mass of tissue = 6 kg

Worked example: Try yourself 3.5.2

DOSE EQUIVALENT

Calculate the dose equivalent (in mSv) from various radioactive sources if the absorbed dose is 1.25 mGy.	
Thinking	Working
The quality factor for each type of radiation can be found in Table 3.5.1. $1 \text{ mGy} = 1 \times 10^{-3} \text{ Gy}$	QF (alpha particles) = 20 QF (beta particles) = 1 QF (gamma rays) = 1

a Calculate the dose equivalent (in mSv) from a radiation source if the absorbed dose is 1.25 mGy and the source is emitting alpha particles.

Thinking

The dose equivalent, $DE = AD \times QF$.
Convert the answer to mSv using
 $1 \text{ mSv} = 1 \times 10^{-3} \text{ Sv}$.

Working

$$\begin{aligned} DE(\alpha) &= 1.25 \times 10^{-3} \times 20 \\ &= 0.025 \text{ Sv} \\ &= 25 \text{ mSv} \end{aligned}$$

b Calculate the dose equivalent (in mSv) from a radiation source if the absorbed dose is 1.25 mGy and the source is emitting beta particles.

Thinking

The dose equivalent, $DE = AD \times QF$.
Convert the answer to mSv.

Working

$$\begin{aligned} DE(\beta) &= 1.25 \times 10^{-3} \times 1 \\ &= 1.25 \times 10^{-3} \\ &= 1.25 \text{ mSv} \end{aligned}$$

c Calculate the dose equivalent (in mSv) from a radiation source if the absorbed dose is 1.25 mGy and the source is emitting gamma rays.

Thinking

The dose equivalent, $DE = AD \times QF$.
Convert the answer to mSv.

Working

$$\begin{aligned} DE(\gamma) &= 1.25 \times 10^{-3} \times 1 \\ &= 1.25 \times 10^{-3} \\ &= 1.25 \text{ mSv} \end{aligned}$$

Worked example: Try yourself 3.5.3

TREATING TUMOURS

A 25g cancer tumour absorbs $5.0 \times 10^{-3} \text{ J}$ of energy from an applied radiation source. Calculate the dose equivalent if the source is an alpha emitter, using information from Table 3.5.1.

Thinking

Convert mass from grams to kg.

Working

$$\begin{aligned} m &= \frac{25}{1000} \\ &= 0.025 \text{ kg} \end{aligned}$$

Calculate the absorbed dose (AD) using the energy and the mass.

$$\begin{aligned} AD &= \frac{E}{m} \\ &= \frac{5 \times 10^{-3}}{0.025} \\ &= 0.2 \text{ Gy} \end{aligned}$$

Calculate the dose equivalent (DE) using the quality factor for alpha particles of 20.

$$\begin{aligned} DE &= AD \times QF \\ &= 0.2 \times 20 \\ &= 4 \text{ Sv} \end{aligned}$$

Section 3.5 Review

KEY QUESTIONS SOLUTIONS

- 1 A. The unit, Gy, indicates that this is the absorbed dose. 1 Gy of alpha radiation will cause 20 times the damage to human tissue than beta or gamma radiation.
- 2 D. The unit, Sv, indicates that this is the dose equivalent. $250 \mu\text{Sv}$ of any of the types of radiation stated will cause the same amount of damage.

- 3 a Absorbed dose, $AD = 200 \mu\text{Gy} = 200 \times 10^{-6} \text{Gy}$
Dose equivalent, $DE = AD \times QF = 200 \times 10^{-6} \times 1 = 200 \times 10^{-6} \text{Sv} = 200 \mu\text{Sv}$
b Absorbed dose, $AD = \frac{E}{m}$
Absorbed energy, $E = AD \times m = 200 \times 10^{-6} \times 80 = 0.016 \text{ J}$
- 4 The amounts given are in Gy so they are the absorbed dose. Use $DE = AD \times QF$ to calculate the equivalent dose for each:
A $DE = 250 \times 10^{-6} \times 1 = 250 \times 10^{-6} \text{Sv} = 250 \mu\text{Sv}$
B $DE = 20 \times 10^{-6} \times 20 = 400 \times 10^{-6} \text{Sv} = 400 \mu\text{Sv}$
C $DE = 50 \times 10^{-6} \times 1 = 50 \times 10^{-6} \text{Sv} = 50 \mu\text{Sv}$
D $DE = 30 \times 10^{-6} \times 10 = 300 \times 10^{-6} \text{Sv} = 300 \mu\text{Sv}$
The highest dose equivalent is $400 \mu\text{Sv}$, which comes from $20 \mu\text{Gy}$ of alpha radiation (B), followed by D, A, and C.
- 5 a The radiation doses are given in μSv so they are dose equivalents (DE).
Convert normal background annual dose to μSv .
 $2 \text{ mSv} = 2 \times 10^3 \mu\text{Sv} = 2000 \mu\text{Sv}$
Number of days $= \frac{2000}{1000} = 2 \text{ days}$
b Number of days in space $= 879 \text{ days}$
Total radiation $= 1000 \mu\text{Sv per day} \times \text{number of days}$
 $= 1000 \times 879$
 $= 879\,000 \mu\text{Sv} = 879 \text{ mSv}$
This equates to 366 mSv per year . This is high but still slightly below the threshold for radiation sickness.
- 6 Number of hours $= \frac{\text{total absorbed dose}}{\text{absorbed dose per hour}}$
 $= \frac{36}{0.40}$
 $= 90 \text{ hours}$
- 7 D. A gamma emitter is needed for the camera, plus it minimises damage to the surrounding tissue as its ability to ionise tissue is low. A short half-life is needed to reduce patient exposure to radiation.

CHAPTER 3 REVIEW

- 20 protons and 25 neutrons ($45 - 20$)
- Cobalt-60 has 27 protons, 33 neutrons ($60 - 27$) and 60 nucleons.
- The atomic and mass numbers of X are both 0, so X is a gamma ray.
- Potassium has $48 - 19 = 29$ neutrons. Figure 3.1.10 shows a minus sign so it emits a beta-minus particle.
- beta minus
 - proton
 - alpha
 - neutron
 - gamma
 - beta plus (positron)
- atomic number $= 5 - 2 = 3$, mass number $= 11 - 4 = 7$, so X is lithium
- atomic number $= (7 + 2) - 8 = 1$, mass number $= (14 + 4) - 17 = 1$, so X is a proton
 - atomic number $= (12 + 1) - 13 = 0$, mass number $= (27 + 1) - 27 = 1$, so Y is a neutron
- $208 = x + 0 \rightarrow x = 208$
 $81 = y - 1 \rightarrow y = 82$
 - $180 = x + 4 \rightarrow x = 176$
 $80 = y + 2 \rightarrow y = 78$
- $18 = a + 0 \rightarrow a = 18$
 $10 = b + 1 \rightarrow b = 9$
 $18 = c + 0 \rightarrow c = 18$
 $9 = d + 1 \rightarrow d = 8$
X has atomic number 9, which is fluorine, F.
Y has atomic number 8, which is oxygen, O.

- 10** atomic number = 12, mass number = 7 – 1 = 6, X is carbon-12
- 11** Electromagnetic forces are balanced by the strong nuclear force acting between all nucleons in close proximity.
- 12** **a** gamma
b gamma
- 13** All types of ionising radiation, including alpha, beta, and gamma.
- 14** The bombarding electrons will be strongly repelled by the electron clouds of the atoms as they are all negatively charged. The small mass of the bombarding electrons also makes them relatively easy to repel compared to, for example, a proton.
- 15** 1 half-life has passed so

$$N = 6.0 \times 10^{14} \times \left(\frac{1}{2}\right)^1 = 3.0 \times 10^{14}$$
- 16** 3 half-lives have passed so

$$N = 5.6 \times 10^{15} \times \left(\frac{1}{2}\right)^3 = 7.0 \times 10^{14}$$
- 17** To have a shorter half-life, the nuclei are decaying at a faster rate, so uranium-235 has a greater activity.
- 18** **a** After one half-life the activity halves to 2 MBq.
b 6 hours
c 18 hours pass = 3 half-lives

$$4.0 \times 10^6 \times \left(\frac{1}{2}\right)^3 = 5.0 \times 10^5 \text{ Bq}$$
- 19** 2 half-lives pass

$$N = 6.0 \times 10^{10} \times \left(\frac{1}{2}\right)^2 = 1.5 \times 10^{10}$$
- 20** The long half-life means that the source will not need to be replaced for many years. The gamma rays have a strong penetrating power so they are able to penetrate the skull and reach the tumour site.
- 21** $DE = AD \times QF = 300 \times 10^{-3} \times 1 = 300 \times 10^{-3} = 300 \text{ mSv}$
- 22** **a** $AD = \frac{E}{m}$ so $E = AD \times m = 5 \times 75 = 375 \text{ J}$
b $DE = AD \times QF = 5 \times 1 = 5 \text{ Sv}$
- 23** **a** Worker works for $45 \times 5 = 225$ days
 Number of X-ray photographs = $225 \times 10 = 2250$ X-rays
 Dose per X-ray = $\frac{7900}{2250} = 3.51 \text{ } \mu\text{Sv}$
b Normal background is 1.5 mSv in Australia = $1500 \text{ } \mu\text{Sv}$
 Number of times greater = $\frac{7900}{1500} = 5.3$ times